3.7 Gate 0: Logic and Sets Examination

The five chapters of *Logic and Proof* and the first three chapters of this volume taught one skill in many forms: reading a mathematical statement exactly, and proving it in sentences that a reader can check line by line. This section is an examination on all of it, and the test of whether *Analysis on the Real Line* will be readable: from its Chapter 1 on, every proof assumes that negating a quantified statement, naming a witness, running an induction and chasing an element through a preimage are routine. A reader who comes to the book by a reading path that begins with *Analysis on the Real Line*, such as a first course in real analysis, takes this Gate first, as an entry test, and goes back only to the sections that the remediation map at the end names. Grading your own work is a skill too; the front matter's page on getting feedback without a teacher describes the loop of attempting, writing, grading against the scheme and only then reading the solution.

3.7.1 Instructions

Algorithm.

- **Time: 2 hours.** Closed book, no notes; paper and pen. Attempt all six problems. The times in brackets are a guide; they add up to 120 minutes.
- Write complete proofs in plain prose, at level 3 of Logic and Proof, Definition 8.3.1. Rule tags are allowed where a problem asks for them.
- Each problem is worth 10 points, 60 in all. Its solution is folded. Open a solution only after the two hours, and **grade before you read it**: apply the marking scheme at the end of the solution to your own script, point by point, and only then read the book's proof.
- **Pass line: 70% overall (42 of 60 points), and no core problem at 0.** The core problems are marked `core`: P1, P2, P4 and P5. They test what *Analysis on the Real Line*, Chapter 1 uses from its first page: quantifiers and their negation, turning a derivation into prose, induction and recursion, sets and functions. P3 and P6 are not core.
- If you fail, follow the remediation map, then wait a week and retake the whole Gate (the same problems) before starting *Analysis on the Real Line*.

The definition drill comes first and is not scored. It is the one retrieval list for those eight chapters: every major definition in them, to be written from memory before the clock starts.

Warning. A proof that reads well is not thereby correct, and grading your own script is where a reader working alone is most easily fooled. Grade the step, not the impression: a marking-scheme line such as "y given as a function of δ" is either on your page or it is not. A script that says "choose y close to 1" in Problem P1 has not given y as a function of δ, however clear the idea, and loses those points.

3.7.2 Definition drill

Quick check. State each of the following **in symbols**, with every quantifier and every bound written out, as the book defines it. Allow about fifteen minutes, before the two hours start.

(Q1) $\Gamma \vdash \psi$.

(Q2) $\Gamma \models \psi$, for formulas of propositional logic.

(Q3) φ is a tautology.

(Q4) The side conditions of $\forall$I and of $\exists$E on the flagged name x_0.

(Q5) $\exists! x\, \varphi$.

(Q6) $A \subseteq B$.

(Q7) $\bigcap_{i \in I} A_i$, for a family $(A_i)_{i \in I}$ of subsets of a set X.

(Q8) A function $f \colon X \to Y$.

(Q9) $f \colon X \to Y$ is injective; f is surjective.

(Q10) The image $f(A)$ and the preimage $f^{-1}(B)$.

(Q11) An equivalence relation on X; a partition of X.

(Q12) A greatest element and a maximal element of $A \subseteq X$, for a partial order $\preceq$ on X.

(Q13) A least upper bound of A in a partially ordered set $(X, \preceq)$.

(Q14) The principle of induction; the well-ordering of $\mathbb{N}$.

(Q15) A is countable.

(Q16) $|A| = |B|$; $|A| \leq |B|$.

(Q17) The axiom of choice.

Solution. The answers are the definitions as the book states them; the label after each answer is where to look when an answer is wrong.

(Q1) For a list $\Gamma = \varphi_1, \ldots, \varphi_n$ ($n \geq 0$): $\Gamma \vdash \psi$ if **there is** a derivation whose premises all belong to Γ and whose last line is ψ (Logic and Proof, Definition 1.2.3).

(Q2) $\Gamma \models \psi$ if for **every** valuation v: if $v(\varphi) = \mathrm{T}$ for every φ in Γ, then $v(\psi) = \mathrm{T}$ (Logic and Proof, Definition 4.5.3).

(Q3) $v(\varphi) = \mathrm{T}$ for **every** valuation v (Logic and Proof, Definition 4.5.1).

(Q4) $\forall$I, closing a flagged box with name x_0 and writing $\forall x\, \varphi$ on line k: at line k, x_0 appears in no premise, in no assumption of a box still open at line k, and not in $\forall x\, \varphi$ (Logic and Proof, Definition 5.4.1). $\exists$E, from $\exists x\, \varphi$ and a flagged box with name x_0 assuming $\varphi[x_0/x]$ and ending in χ, writing χ on line k: x_0 does not appear in χ, in $\exists x\, \varphi$, in any premise, or in any assumption of a box still open at line k (Logic and Proof, Definition 5.6.1). "Appears" means written anywhere, after a quantifier symbol included.

(Q5) $\exists x\, (\varphi \wedge \forall y\, (\varphi[y/x] \to y = x))$, where y is a variable other than x that does not appear in φ (Logic and Proof, Definition 5.8.1).

(Q6) $\forall x\, (x \in A \to x \in B)$ (Definition 1.1.2).

(Q7) $\bigcap_{i \in I} A_i \coloneqq \{x \in X : \forall i \in I\ (x \in A_i)\}$; for $I = \emptyset$ it is X (Definition 1.2.4).

(Q8) A relation $G \subseteq X \times Y$, the graph, with $\forall x \in X\ \exists!\, y \in Y\ (x, y) \in G$, together with the codomain Y: formally $f = (G, Y)$ (Definition 1.4.1).

(Q9) Injective: $\forall x_1 \in X\ \forall x_2 \in X\ (f(x_1) = f(x_2) \to x_1 = x_2)$ (Definition 1.4.3). Surjective: $\forall y \in Y\ \exists x \in X\ f(x) = y$ (Definition 1.4.4).

(Q10) For $A \subseteq X$: $y \in f(A) \iff y \in Y \wedge \exists a \in A\ f(a) = y$ (Definition 1.6.1). For $B \subseteq Y$: $x \in f^{-1}(B) \iff x \in X \wedge f(x) \in B$ (Definition 1.6.2).

(Q11) A relation $\sim$ on X with (Eq1) $\forall x \in X\ x \sim x$; (Eq2) $\forall x, y \in X\ (x \sim y \to y \sim x)$; (Eq3) $\forall x, y, z \in X\ (x \sim y \wedge y \sim z \to x \sim z)$ (Definition 2.5.1). A set $\mathcal{P}$ of subsets of X with (P1) $\forall P \in \mathcal{P}\ P \neq \emptyset$; (P2) $\forall P, Q \in \mathcal{P}\ (P \neq Q \to P \cap Q = \emptyset)$; (P3) $\bigcup \mathcal{P} = X$ (Definition 2.5.3).

(Q12) g is a greatest element: $g \in A \wedge \forall a \in A\ a \preceq g$ (Definition 2.6.3). m is a maximal element: $m \in A \wedge \neg\exists a \in A\ m \prec a$, where $m \prec a$ means $m \preceq a \wedge m \neq a$ (Definition 2.6.4).

(Q13) $s \in X$ with (S1) $\forall a \in A\ a \preceq s$, and (S2) $\forall u \in X\ ((\forall a \in A\ a \preceq u) \to s \preceq u)$ (Definition 2.6.5).

(Q14) For every property P of natural numbers: if $P(0)$ and $\forall n \in \mathbb{N}\ (P(n) \to P(n+1))$, then $\forall n \in \mathbb{N}\ P(n)$ (Logic and Proof, Axiom 4.2.1). Every non-empty $S \subseteq \mathbb{N}$ has a least element: $S \neq \emptyset \to \exists m \in S\ \forall s \in S\ m \leq s$ (Theorem 2.1.3).

(Q15) $A = \emptyset$, or there is $f\colon \mathbb{N} \to A$ with $\forall a \in A\ \exists n \in \mathbb{N}\ f(n) = a$ (Definition 3.2.1).

(Q16) $|A| = |B|$ if there is a bijection $A \to B$ (Definition 3.1.1); $|A| \leq |B|$ if there is an injection $A \to B$ (Definition 3.5.1).

(Q17) For every set I and every family $(A_i)_{i \in I}$ of sets with $\forall i \in I\ A_i \neq \emptyset$, there is $c\colon I \to \bigcup_{i \in I} A_i$ with $\forall i \in I\ c(i) \in A_i$ (Axiom 3.3.1).

Count an item right only if every quantifier, every bound and the order of the quantifiers match. The commonest slips are a missing "$\in X$" bound in (Q9), "$\exists y$" for "$\exists!\, y$" in (Q8), and, in (Q12), "$\forall a \in A\ a \preceq m$" for maximal, which is the definition of greatest.

3.7.3 The problems

Exercise. P1: core

(20 minutes.) Say that a function $f\colon \mathbb{R} \to \mathbb{R}$ has **property L** if for every $x \in \mathbb{R}$ there is $\delta > 0$ such that for every $y \in \mathbb{R}$ with $|y - x| < \delta$, we have $f(y) \geq f(x)$.

(a) Write property L in symbols, with bounded quantifiers.

(b) Write its negation with no $\neg$ in front of a quantifier, a connective or a parenthesis.

(c) Prove that $f(x) = x^2$ does not have property L.

(d) Prove that the function $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x) = 1$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$ (the indicator of $(0, \infty)$, with codomain $\mathbb{R}$) has property L, and that it does **not** have the property obtained from L by moving "there is $\delta > 0$" to the front.

Solution. (a) Property L is

$$\forall x \in \mathbb{R}\ \exists \delta > 0\ \forall y \in \mathbb{R}\ \big(|y - x| < \delta \to f(y) \geq f(x)\big).$$

(b) Push the negation inward one symbol at a time, as in the procedure of *Logic and Proof*, Section 4.9 ("Negating a definition, mechanically"): Logic and Proof, Proposition 6.2.3 three times, the bound of each quantifier kept, and Logic and Proof, Proposition 6.2.2 once. Write

α for $|y-x|<\delta$ and β for $f(y)\geq f(x)$.

$$\begin{aligned}&\neg\,\forall x\in\mathbb{R}\;\exists\delta>0\;\forall y\in\mathbb{R}\;(\alpha\to\beta)\\ \dashv\vdash\;&\exists x\in\mathbb{R}\;\forall\delta>0\;\neg\,\forall y\in\mathbb{R}\;(\alpha\to\beta)\\ \dashv\vdash\;&\exists x\in\mathbb{R}\;\forall\delta>0\;\exists y\in\mathbb{R}\;\neg(\alpha\to\beta)\\ \dashv\vdash\;&\exists x\in\mathbb{R}\;\forall\delta>0\;\exists y\in\mathbb{R}\;(\alpha\wedge\neg\beta).\end{aligned}$$

The first step does two moves at once ($\neg\forall$ to $\exists\neg$, then $\neg\exists$ to $\forall\neg$). After the chain, $\neg\beta$, that is $\neg(f(y)\geq f(x))$, is replaced by $f(y)<f(x)$, a fact about the order of $\mathbb{R}$ and not a law of logic. The negation of L is therefore

$$\exists x\in\mathbb{R}\;\forall\delta>0\;\exists y\in\mathbb{R}\;\big(|y-x|<\delta\wedge f(y)<f(x)\big).$$

Note what stays: the bound $\delta>0$ is kept, and only the quantifier in front of it changes; "$\exists\delta\leq 0$" would be the error of negating the bound. The implication becomes a conjunction, never another implication.

(c) We prove the negation (b) for $f(x)=x^2$, with the witness $x=1$. Let $\delta>0$ be arbitrary, and put

$$y\coloneqq 1-\tfrac{1}{2}\min\{\delta,1\}.$$

Since $0<\min\{\delta,1\}\leq 1$, we have $\frac{1}{2}\leq y<1$. Then $|y-1|=\frac{1}{2}\min\{\delta,1\}\leq\frac{1}{2}\delta<\delta$, and, as $0<y<1$, $y^2=y\cdot y<y\cdot 1<1=f(1)$. So $|y-1|<\delta$ and $f(y)<f(1)$. Since $\delta>0$ was arbitrary, the negation of L holds at $x=1$, and x^2 does not have property L.

The order of the choices is the order of the quantifiers in (b): x is fixed before δ, and y is a function of δ. The minimum with 1 is needed: $y\coloneqq 1-\delta/2$ fails for large δ, since $\delta=4$ gives $y=-1$ and $y^2=1\not<1$.

(d) Let $f\colon\mathbb{R}\to\mathbb{R}$ be the function of (d): $f(x)=1$ for $x>0$ and $f(x)=0$ for $x\leq 0$. Let $x\in\mathbb{R}$.

Case 1: $x>0$. Take $\delta\coloneqq x$, which is positive. If $|y-x|<x$, then $y>x-x=0$, so $f(y)=1=f(x)$, and $f(y)\geq f(x)$.

Case 2: $x\leq 0$. Take $\delta\coloneqq 1$. For every y, $f(y)\in\{0,1\}$, so $f(y)\geq 0=f(x)$.

So f has property L. In Case 1, δ **depends on** x: points close to 0 need small δ.

The swapped property is $\exists\delta>0\;\forall x\in\mathbb{R}\;\forall y\in\mathbb{R}\;(|y-x|<\delta\to f(y)\geq f(x))$. We prove its negation, $\forall\delta>0\;\exists x\in\mathbb{R}\;\exists y\in\mathbb{R}\;(|y-x|<\delta\wedge f(y)<f(x))$. Let $\delta>0$, and take $x\coloneqq\delta/4$ and $y\coloneqq-\delta/4$. Then $|y-x|=\delta/2<\delta$, while $f(y)=0<1=f(x)$, since $y<0<x$. So no single δ serves every x. Here x depends on δ, which is legitimate, because δ is now fixed first.

Marking scheme. (10 points) - 2: (a) correct, with $\forall x\in\mathbb{R}$, $\exists\delta>0$, $\forall y\in\mathbb{R}$ in this order and the condition $|y-x|<\delta$ as the hypothesis of an implication (or as a bound $\forall y\in\mathbb{R}$ with $|y-x|<\delta$). - 3: (b) correct: the bounds kept ($\forall\delta>0$, not $\exists\delta\leq 0$), every quantifier switched, and the implication negated to a conjunction. - 2: (c) a witness x fixed **before** δ, and a y given as a function of δ that works for **every** $\delta>0$ (any correct y, such as $1-\frac{1}{2}\min\{\delta,1\}$ or $1-\min\{\delta/2,\frac{1}{2}\}$, earns the points). - 3: (d) 2 for the case split proving L, with a δ for each x; 1 for refuting the swap, with x and y depending on δ. Fatal errors (0 for the step): a negation that keeps "$\to$"; in (c), y chosen before δ or for one particular δ; in the first half of (d), a single δ claimed for all x.

Exercise. P2: core

(20 minutes.) Here is a derivation of

$$\forall z\,\exists y\,G(y,z),\ \forall y\,\exists x\,F(x,y),$$
$$\forall x\,\forall y\,\forall z\,((F(x,y)\wedge G(y,z))\to H(x,z))\vdash \forall z\,\exists x\,H(x,z),$$

with four justifications left blank, marked (i)–(iv).

1		$\forall z\,\exists y\,G(y,z)$	
2		$\forall y\,\exists x\,F(x,y)$	
3		$\forall x\,\forall y\,\forall z\,((F(x,y)\wedge G(y,z))\to H(x,z))$	
4	z_0	$\exists y\,G(y,z_0)$	$\forall$E 1
5	y_0	$G(y_0,z_0)$	
6		$\exists x\,F(x,y_0)$	(i)
7	x_0	$F(x_0,y_0)$	
8		$\forall y\,\forall z\,((F(x_0,y)\wedge G(y,z))\to H(x_0,z))$	$\forall$E 3
9		$\forall z\,((F(x_0,y_0)\wedge G(y_0,z))\to H(x_0,z))$	$\forall$E 8
10		$(F(x_0,y_0)\wedge G(y_0,z_0))\to H(x_0,z_0)$	$\forall$E 9
11		$F(x_0,y_0)\wedge G(y_0,z_0)$	$\wedge$I 7, 5
12		$H(x_0,z_0)$	$\to$E 10, 11
13		$\exists x\,H(x,z_0)$	(ii)
14		$\exists x\,H(x,z_0)$	(iii)
15		$\exists x\,H(x,z_0)$	$\exists$E 4, 5–14
16		$\forall z\,\exists x\,H(x,z)$	(iv)

(a) Fill in the justifications (i)–(iv): the rule and the cited lines.

(b) Explain why line 13 sits inside both of the boxes that begin at lines 5 and 7, and why each of these two boxes may close where it does.

(c) Read $F(x,y)$ as "$f(x)=y$", $G(y,z)$ as "$g(y)=z$" and $H(x,z)$ as "$g(f(x))=z$", for functions $f\colon X\to Y$ and $g\colon Y\to Z$. Write the plain-prose proof that this derivation encodes, that the composite of two surjections is surjective, with one sentence for each box.

Solution. (a) (i) $\forall$E 2, with the term y_0 for y. (ii) $\exists$I 12, with the witness x_0 for x in $H(x,z_0)$. (iii) $\exists$E 6, 7–13: the existential of line 6 and the box of x_0. (iv) $\forall$I 4–15: the flagged box of z_0, whose range runs from its first line, line 4, to its last line, line 15 (Logic and Proof, Definition 5.4.1).

(b) Line 12, $H(x_0,z_0)$, contains the witness x_0. A box of $\exists$E may close only on a last line in which its own name does not appear (Logic and Proof, Definition 5.6.1). So the box of x_0 cannot close on line 12; the $\exists$I of line 13 replaces x_0 by a bound variable first, and it

must come **before** that box closes, that is, inside it, and therefore inside the box of y_0 that contains it.

The box of x_0 closes at line 14. Its name x_0 does not appear in the last line $\exists x\, H(x, z_0)$, nor in the existential $\exists x\, F(x, y_0)$ of line 6, nor in the premises 1–3 (where only the bound variable x is written), nor in the one open assumption at line 14, $G(y_0, z_0)$ on line 5. The box of y_0 closes at line 15. Its name y_0 does not appear in $\exists x\, H(x, z_0)$, nor in $\exists y\, G(y, z_0)$ of line 4, nor in the premises; the only box still open at line 15 is the flagged box of z_0, which has no assumption. Of the four clauses, the last-line clause is the one that shapes this derivation: the other three hold for any fresh names, while the last line had to be prepared by line 13. The names x_0 and y_0 may not leave their boxes because each names one witness, about which only its box's assumption is known; outside, it would be treated as arbitrary, which is error (N3) of Logic and Proof, Example (What is not allowed). The name z_0 is different: it may appear in lines 14 and 15, because its box is closed by $\forall$I at line 16, where it no longer appears.

(c) Let $f\colon X \to Y$ and $g\colon Y \to Z$ be surjective, and let $z \in Z$ be arbitrary (the box of z_0). Since g is surjective, there is $y \in Y$ with $g(y) = z$; fix one (the box of y_0, lines 5–14, eliminating line 4). Since f is surjective, there is $x \in X$ with $f(x) = y$; fix one (the box of x_0, lines 7–13, eliminating line 6). Then $g(f(x)) = g(y) = z$, so z is a value of $g \circ f$ (lines 8–13; premise 3 is the substitution of equals that the equation chain performs). Since z was arbitrary, $g \circ f$ is surjective, which is the surjectivity half of Theorem 1.4.4. The derivation leaves the bounds $z \in Z$, $y \in Y$, $x \in X$ implicit; the prose restores them.

Marking scheme. (10 points) - 4: (a), one point per blank: the $\forall$E with its line and instance term; the $\exists$I with its line; the inner $\exists$E citing line 6 and the box 7–13; the final $\forall$I citing the box 4–15. - 2: (b) the fresh-name condition of $\exists$E stated with all its clauses, in terms of where the name **appears**, for each box's own name; and why x_0 and y_0 may not leave their boxes (so the $\exists$I must come first). - 4: (c) "let $z \in Z$", "choose y with $g(y) = z$", "choose x with $f(x) = y$", "then $g(f(x)) = z$", in this order, each sentence carrying its box. Fatal errors (0 for the step): "let x be arbitrary" for an $\exists$E; prose in which y is chosen before z is fixed.

Exercise. P3

(20 minutes.) For each argument below, answer three questions. Is the conclusion true? Does the argument prove it? If not, which line fails, which entry of Logic and Proof, Example (What is not allowed) does it fall under, and what is a counterexample **to the illegal step** (not merely to the conclusion)?

(a) *Claim:* $n^2 + n$ is even for every $n \in \mathbb{N}$. *Proof:* $0^2 + 0 = 0$ is even. Assume that $n^2 + n$ is even for all $n \in \mathbb{N}$. Then $(n+1)^2 + (n+1) = (n^2 + n) + 2(n+1)$ is even. By induction, the claim holds.

(b) *Claim:* if $A_n \subseteq \mathbb{R}$ is bounded above for every $n \in \mathbb{N}$, then $\bigcup_{n\in\mathbb{N}} A_n$ is bounded above. *Proof:* For each n, let M_n be an upper bound of A_n. Let M be a number larger than every M_n. Then M is an upper bound of the union.

Solution. *(a)* **The conclusion is true.** Let $n \in \mathbb{N}$. By Theorem 2.2.3 with divisor 2, $n = 2q + r$ with $q \in \mathbb{Z}$ and $r \in \{0, 1\}$. If $r = 0$, then $n^2 + n = n(n+1) = 2\big(q(n+1)\big)$; if $r = 1$, then $n + 1 = 2(q+1)$ and $n^2 + n = 2\big(n(q+1)\big)$. In both cases the factor in brackets

is an integer (Proposition 2.1.2), so $2 \mid n^2 + n$ (Definition 2.2.1).

The argument does not prove it. The failing line is "Assume that $n^2 + n$ is even for all $n \in \mathbb{N}$". That assumption is the claim itself, so everything after it is circular: entry (N9) of Logic and Proof, Example (What is not allowed), citing a line that is not available, here the goal. The step of an induction is a $\to$I box inside a $\forall$I box: fix one arbitrary $n \in \mathbb{N}$, assume that $n^2 + n$ is even **for this** n, and derive that $(n+1)^2 + (n+1)$ is even. With that one line changed, the rest of the argument is correct: the identity holds, since both sides equal $n^2 + 3n + 2$, and a sum of two even integers is even ($2a + 2b = 2(a + b)$).

A counterexample to the illegal step. The argument infers $\forall n\, P(n)$ from $P(0)$ and from "if $P(n)$ for all n, then $P(n+1)$ for all n". Take $P(n)$: "$n = 0$". Then $P(0)$ holds, and the second premise holds because its hypothesis "$P(n)$ for all n" is false ($P(1)$ fails), so the implication is vacuously true. Yet $\forall n\, P(n)$ is false. The conclusion of (a) is true, so it has no counterexample; the step has one.

(b) **The conclusion is false.** Take $A_n := \{n\}$. Each A_n is bounded above, by n. The union is $\mathbb{N}$, and $\mathbb{N}$ is not bounded above in $\mathbb{R}$: for every real M there is a natural number larger than M (a school fact here; *Analysis on the Real Line*, Chapter 1 proves it from its axiom).

The failing line is "Let M be a number larger than every M_n". From "for every n there is a number larger than M_n" it passes to "there is a number larger than M_n for every n": from $\forall n\, \exists M$ to $\exists M\, \forall n$. This is entry (N2) of Logic and Proof, Example (What is not allowed), a witness that depends on n presented as uniform. It is the converse of Logic and Proof, Proposition 5.6.2, and the converse is not even valid: Logic and Proof, Example (The swap of $\exists$ and $\forall$) gives a counter-model. *A counterexample to the step:* $M_n := n$. For each n there is a number larger than M_n, namely $n + 1$, and no real number is larger than every n.

The line before it is not the error. Choosing an upper bound M_n for every n at once, with no rule, is a use of the axiom of choice (Axiom 3.3.1), and the book allows it, flagged. And for **finitely** many indices the bad line becomes correct: $\max\{M_0, \ldots, M_k\} + 1$ exists by Proposition 2.3.8, so a union of finitely many sets bounded above is bounded above.

Marking scheme. (10 points) - (a), 5 points. 1: says the conclusion is true, with a reason (a proof or a correct induction). 2: points to the line "assume ... for all n". 1: names the error, circularity, (N9). 1: gives a counterexample to the step, not to the conclusion. - (b), 5 points. 1: says the conclusion is false, with a witness such as $A_n = \{n\}$. 2: points to the line "let M be larger than every M_n". 1: names the quantifier swap $\forall n\, \exists M$ to $\exists M\, \forall n$, or (N2). 1: gives a counterexample to the step, not to the conclusion, such as $M_n = n$. Fatal errors (0 for the part): in (a), calling the argument correct; in (b), locating the error in the choice of the M_n.

Exercise. P4: core

(20 minutes.) Let $a_0 = 0$, $a_1 = 1$ and $a_{n+2} = 3a_{n+1} - 2a_n$ for $n \in \mathbb{N}$.

(a) Say which result guarantees that these equations define a sequence.

(b) Prove that $a_n = 2^n - 1$ for every $n \in \mathbb{N}$.

(c) Suppose instead that $a_1 = 2$, with a_0 and the rule unchanged. Compute a_2 and a_3, and say exactly which line of your proof in (b) fails.

Solution. (a) Two-step recursion, Corollary 2.3.3, with $X = \mathbb{R}$, $a = 0$, $b = 1$ and $h(n, x, y) := 3y - 2x$: there is exactly one sequence $(a_n)_{n\in\mathbb{N}}$ of reals with $a_0 = 0$, $a_1 = 1$ and $a_{n+2} = 3a_{n+1} - 2a_n$ for every $n \in \mathbb{N}$.

(b) The rule looks two steps back, so the induction carries two values at once. For $n \in \mathbb{N}$ let $Q(n)$ be the statement

$$a_n = 2^n - 1 \quad \text{and} \quad a_{n+1} = 2^{n+1} - 1.$$

Base case. $Q(0)$ says $a_0 = 2^0 - 1 = 0$ and $a_1 = 2^1 - 1 = 1$, and both hold.

Step. Let $n \in \mathbb{N}$ and suppose $Q(n)$. The second half of $Q(n)$ is the first half of $Q(n+1)$. For the second half, by the rule and both halves of $Q(n)$,

$$\begin{aligned} a_{n+2} &= 3a_{n+1} - 2a_n = 3(2^{n+1} - 1) - 2(2^n - 1) \\ &= 3 \cdot 2^{n+1} - 2^{n+1} - 1 = 2 \cdot 2^{n+1} - 1 = 2^{n+2} - 1, \end{aligned}$$

where $2 \cdot 2^n = 2^{n+1}$ and $2 \cdot 2^{n+1} = 2^{n+2}$ by Definition 2.3.2. So $Q(n+1)$ holds. By Logic and Proof, Axiom 4.2.1, $Q(n)$ holds for every $n \in \mathbb{N}$, and in particular $a_n = 2^n - 1$ for every $n \in \mathbb{N}$. This is the pair induction of the two-step example in Section 2.3, after Definition 2.3.2.

Alternatively, by strong induction (Logic and Proof, Theorem 4.2.2): let $n \in \mathbb{N}$ and suppose $a_k = 2^k - 1$ for every $k < n$. If $n = 0$ or $n = 1$, check the value directly. If $n \geq 2$, then $n - 1$ and $n - 2$ are natural numbers (Lemma 2.1.1 (c), twice) smaller than n, so the hypothesis applies to both, and the same computation with $n - 2$ in place of n gives $a_n = 2^n - 1$. The two values $n = 0, 1$ are exactly where the hypothesis would be used out of range.

(c) Now $a_2 = 3 \cdot 2 - 2 \cdot 0 = 6$ and $a_3 = 3 \cdot 6 - 2 \cdot 2 = 14$, while $2^2 - 1 = 3$ and $2^3 - 1 = 7$. The failing line is the base case at $n = 1$: in the proof of (b), the half "$a_1 = 2^1 - 1$" of $Q(0)$, since now $a_1 = 2 \neq 1$. The check $a_0 = 0$ still holds, and the step is still a valid implication; but every value from a_2 on is computed from a_1, so the failure propagates. (The same induction proves $a_n = 2^{n+1} - 2$ for the new sequence.)

Marking scheme. (10 points) - 1: (a) names two-step recursion (Corollary 2.3.3), or Theorem 2.3.1 applied to pairs. - 2: (b) states a form of induction under which both a_0 and a_1 are checked: two base cases for strong induction, or the base $Q(0)$ of a pair statement. - 3: (b) the step's algebra, $3(2^{n+1} - 1) - 2(2^n - 1) = 2^{n+2} - 1$ (or the same with shifted indices). - 1: (b) the induction hypothesis used only for indices where it is available (no a_{-1}; in strong induction, $n \geq 2$ in the step). - 1: (b) the conclusion, citing the induction principle used. - 2: (c) $a_2 = 6$, $a_3 = 14$ (1), and the failing line identified as the check of a_1 (1). Fatal errors (0 for the step): only a_0 checked, so a_1 is never verified; an induction hypothesis "for all n", entry (N9) of Logic and Proof, Example (What is not allowed).

Exercise. P5: core

(20 minutes.) Let $f\colon X \to Y$ be a function.

(a) For a family $(B_i)_{i\in I}$ of subsets of Y, prove that

$$f^{-1}\Big(Y \setminus \bigcup_{i\in I} B_i\Big) = \bigcap_{i\in I} (X \setminus f^{-1}(B_i)),$$

with the intersection taken in X.

(b) Prove that $f(X \setminus A) \subseteq Y \setminus f(A)$ for **every** $A \subseteq X$ if and only if f is injective.

Solution. (a) Put $B \coloneqq \bigcup_{i \in I} B_i$, a subset of Y since every B_i is. Then

$$f^{-1}(Y \setminus B) = X \setminus f^{-1}(B) = X \setminus \bigcup_{i \in I} f^{-1}(B_i)$$
$$= \bigcap_{i \in I} (X \setminus f^{-1}(B_i)).$$

The first equality is Theorem 1.6.1 (c), the second is Theorem 1.6.1 (a), and the third is Theorem 1.2.3, applied to the family $\left(f^{-1}(B_i)\right)_{i \in I}$ of subsets of X, with the intersection taken in X. Nothing excludes $I = \emptyset$: then both sides are X.

Alternatively, element by element. For every x: $x \in f^{-1}(Y \setminus B)$ iff $x \in X$ and $f(x) \in Y \setminus B$, iff $x \in X$ and $f(x) \notin B$ (as $f(x) \in Y$ anyway), iff $x \in X$ and there is no $i \in I$ with $f(x) \in B_i$, iff $x \in X$ and for every $i \in I$, $x \notin f^{-1}(B_i)$, iff $x \in X$ and for every $i \in I$, $x \in X \setminus f^{-1}(B_i)$, iff $x \in \bigcap_{i \in I}(X \setminus f^{-1}(B_i))$. The steps use Definition 1.6.2, Definition 1.2.4, Definition 1.2.1 and Logic and Proof, Proposition 6.2.3 (b).

(b) ($\Leftarrow$) Suppose f is injective, and let $A \subseteq X$ and $y \in f(X \setminus A)$. By Definition 1.6.1 there is $x \in X \setminus A$ with $f(x) = y$; in particular $y \in Y$. Suppose, for contradiction, that $y \in f(A)$. Then there is $a \in A$ with $f(a) = y$. So $f(x) = f(a)$, and injectivity gives $x = a \in A$, contradicting $x \notin A$. Hence $y \notin f(A)$, so $y \in Y \setminus f(A)$. As y was arbitrary, $f(X \setminus A) \subseteq Y \setminus f(A)$, and A was arbitrary.

($\Rightarrow$) By contrapositive, as in Logic and Proof, Example (Even square, even number, by contraposition). Suppose f is not injective: by Logic and Proof, Proposition 6.2.3 (a), twice, and Logic and Proof, Proposition 6.2.2 there are $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$ and $x_1 \neq x_2$. Take $A \coloneqq \{x_1\}$. Then $x_2 \in X \setminus A$, so $f(x_2) \in f(X \setminus A)$. But $f(x_2) = f(x_1) \in f(A)$, so $f(x_2) \notin Y \setminus f(A)$. So the inclusion fails for this A, and it does not hold for every $A \subseteq X$.

The two inclusions hide one witness each: $y \in f(X \setminus A)$ hands us an x, and $y \in f(A)$ hands us an a. The proof works by comparing the two, which is what injectivity is for. Compare Theorem 1.6.4 (b), which characterizes injectivity through intersections in the same way.

Marking scheme. (10 points) - 4: (a) by either route. For the first, each cited result must match its use (which part of Theorem 1.6.1, and De Morgan in X); for the second, each equivalence justified by a definition. A proof of the two inclusions separately earns the same marks. - 3: (b) ($\Leftarrow$), with the witness $x \in X \setminus A$ of $y \in f(X \setminus A)$ and the witness $a \in A$ of $y \in f(A)$ both named, and injectivity applied to them. - 3: (b) ($\Rightarrow$), by contrapositive or contradiction, with a correct set A built from the two points with the same value. Fatal errors (0 for the step): treating $y \in f(A)$ as "$x \in A$" for the x already in hand (a skipped $\exists$E); proving ($\Rightarrow$) from the inclusion for one given A only (the hypothesis "for every A" dropped; choosing one A where it fails, as above, is correct), or with "for every A" read as "for some A".

Exercise. P6

(20 minutes.) Let S be the set of sequences $(a_n)_{n \in \mathbb{N}}$ of natural numbers, indexed from 0, with $a_n \leq n$ for every $n \in \mathbb{N}$; a sequence is a function $\mathbb{N} \to \mathbb{N}$, so $S \subseteq \mathbb{N}^{\mathbb{N}}$.

(a) Prove that S is uncountable.

(b) A sequence (a_n) is **eventually constant** if there is $N \in \mathbb{N}$ with $a_n = a_N$ for every $n \geq N$. Let $E \subseteq S$ consist of the sequences in S that are eventually constant. Prove that E is countable, and explain why running your construction from (a) on a list of E produces no contradiction.

Solution. (a) The obvious diagonal, $g(n) \coloneqq f_n(n) + 1$, leaves S: already $g(0) \geq 1 > 0$. Worse, place 0 offers no room at all, since every sequence in S has $a_0 = 0$. So the diagonal is shifted by one place: the n-th sequence of the list is made to differ at place $n + 1$, where there are at least two values to choose from.

Let $f\colon \mathbb{N} \to S$ be an arbitrary function, and write $f_n \coloneqq f(n)$. Define $g\colon \mathbb{N} \to \mathbb{N}$ by $g(0) \coloneqq 0$ and, for $n \in \mathbb{N}$,

$$g(n+1) \coloneqq \begin{cases} 0 & \text{if } f_n(n+1) \neq 0, \\ 1 & \text{if } f_n(n+1) = 0. \end{cases}$$

This defines g on all of $\mathbb{N}$, since every natural number is 0 or $n + 1$ for exactly one $n \in \mathbb{N}$ (Lemma 2.1.1 (c)). Then $g \in S$: $g(0) = 0 \leq 0$, and $g(n + 1) \leq 1 \leq n + 1$. For every $n \in \mathbb{N}$, $g(n + 1) \neq f_n(n + 1)$ by the two cases, so $g \neq f_n$. Hence g is not a value of f, and f is not surjective. Since f was arbitrary, there is no surjection $\mathbb{N} \to S$. And $S \neq \emptyset$, since the zero sequence lies in S. By Definition 3.2.1, S is not countable.

(b) For $N \in \mathbb{N}$ let E_N be the set of sequences $(a_n) \in S$ with $a_n = a_N$ for every $n \geq N$. By the definition of eventually constant, $E = \bigcup_{N \in \mathbb{N}} E_N$.

Each E_N is countable. For $N = 0$, a sequence in E_0 is constant with value $a_0 \leq 0$, so E_0 has one element, the zero sequence; a constant map $\mathbb{N} \to E_0$ is onto, and E_0 is countable. Let $N \geq 1$, and define $\pi\colon E_N \to \mathbb{N}^{N+1}$ by $\pi(a) \coloneqq (a_0, a_1, \ldots, a_N)$, an $(N + 1)$-tuple. It is injective: if $\pi(a) = \pi(b)$, then $a_k = b_k$ for $k \leq N$, by the characteristic property of tuples (Section 1.3), and for $n \geq N$, $a_n = a_N = b_N = b_n$; so $a = b$ (Proposition 1.4.1). The set $\mathbb{N}^{N+1}$ is countable by Corollary 3.2.7, as $N + 1 \geq 2$, so by Theorem 3.2.2, (i) $\Rightarrow$ (ii), there is an injection $j\colon \mathbb{N}^{N+1} \to \mathbb{N}$. Then $j \circ \pi\colon E_N \to \mathbb{N}$ is injective (Theorem 1.4.4), and E_N is countable by (ii) $\Rightarrow$ (i).

So E is countable, as a union of countably many countable sets, by Theorem 3.3.3. That theorem uses the axiom of choice once, to pick a list of each E_N. Here the choice can be avoided (compare Example (Unions that come with their lists); see the marking scheme). The solution does not need this; citing Theorem 3.3.3 is a complete proof.

In fact each E_N is finite: $a_0, a_1, \ldots, a_N$ can take $1, 2, \ldots, N + 1$ values, so E_N has $(N + 1)!$ elements (Definition 2.3.1), which is $1, 2, 6, 24, 120$ for $N = 0, \ldots, 4$. This count rests on Exercise (C2: Counting products and functions) and an induction on the number of factors, and it earns the same marks.

Why there is no contradiction. The set E is countable and non-empty, so it has a list: a surjection $e\colon \mathbb{N} \to E$. The construction of (a) applies to e, since e is in particular a function $\mathbb{N} \to S$, and produces $g \in S$ with $g \neq e(n)$ for every n. Since every element of E is some $e(n)$, this shows $g \notin E$: the diagonal sequence is not eventually constant. That is all it shows. A diagonal argument proves that a given list misses something in the set the diagonal element lives in, here S; it contradicts the countability of E only if the diagonal element lies in E, and it does not.

Marking scheme. (10 points) - 1: (a) starts from an arbitrary list, a function $\mathbb{N} \to S$, not

a particular one. - 2: (a) defines a diagonal element that lies in S, with the problem at place 0 handled. - 2: (a) shows that it differs from every f_n, at a named place. - 1: (a) concludes, with the form "countable means empty or the image of $\mathbb{N}$" handled ($S \neq \emptyset$). - 2: (b) writes E as a countable union of countable sets, with a reason why each piece is countable (an injection into $\mathbb{N}$, a surjection from $\mathbb{N}$, or finiteness via Exercise (C2: Counting products and functions)). Alternative route, same 2 points: an injection of E into the finite sequences of Exercise (B2: Finite sequences), $a \mapsto (k \mapsto a_{k-1})$ on $\{1, \ldots, N+1\}$ with N least such that $a \in E_N$, with injectivity checked and Theorem 3.2.2 cited. - 2: (b) explains that the diagonal element of a list of E lies in S but not in E, so it contradicts nothing. Fatal errors (0 for the step): a diagonal element not in S, such as $g(n) \coloneqq f_n(n) + 1$, which already has $g(0) \geq 1 > 0$; proving (a) for one particular list.

3.7.4 Remediation map

Add up your points. A pass is 42 of 60 with no core problem (P1, P2, P4, P5) at 0. For each problem where you lost more than 3 points, reread the sections in its row, redo their A exercises and one C exercise. Then retake that problem if you passed, or the whole Gate after a week if you failed. A wrong answer in the definition drill sends you to the label printed after its answer.

Problem	What it tests	Sections to reread
P1	quantifiers, bounds and their negation; the order of choices	*Logic and Proof*, Section 4.1 (Logic and Proof, Definition 5.1.1); *Logic and Proof*, Section 4.9 (Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.2, Logic and Proof, Proposition 6.2.3); *Logic and Proof*, Section 4.10 (Logic and Proof, Example (Uniformly continuous implies continuous)); *Logic and Proof*, Section 5.2 (Logic and Proof, Example (A constant sequence))
P2	the quantifier rules and their side conditions; a derivation read as prose	*Logic and Proof*, Sections 4.3–4.6 (Logic and Proof, Definition 5.3.1, Logic and Proof, Definition 5.4.1, Logic and Proof, Definition 5.5.1, Logic and Proof, Definition 5.6.1); *Logic and Proof*, Sections 5.2–5.3 (Logic and Proof, Example (The reciprocals tend to zero), Logic and Proof, Definition 8.3.1); Section 1.4 (Theorem 1.4.4)

Problem	What it tests	Sections to reread
P3	checking a proof; is the conclusion true, and does the argument prove it	*Logic and Proof*, Section 5.6 (Logic and Proof, Example (What is not allowed)); *Logic and Proof*, Section 3.2 (Logic and Proof, Axiom 4.2.1); *Logic and Proof*, Section 4.6 (Logic and Proof, Proposition 5.6.2); *Logic and Proof*, Section 4.11 (Logic and Proof, Example (The swap of $\exists$ and $\forall$)); Section 2.2 (Theorem 2.2.1, and "How induction proofs go wrong", Example (All Numbers Are Equal))
P4	recursion and induction with two starting values	*Logic and Proof*, Section 3.2 (Logic and Proof, Theorem 4.2.2); Section 2.2 (Theorem 2.2.2); Section 2.3 (Corollary 2.3.3, and the two-step example after Definition 2.3.2)
P5	images and preimages, element by element	Section 1.2 (Theorem 1.2.3); Section 1.6 (Theorem 1.6.1, Theorem 1.6.4)
P6	countable sets and the diagonal argument	Section 3.2 (Definition 3.2.1); Section 3.3 (Theorem 3.3.3); Section 3.4 (Theorem 3.4.1, Theorem 3.4.2)